DAY 2

Text Representation

TF-IDF & Embeddings

From clean text to numbers — and from numbers to meaning.

IMDb Clean Reviews

Week 8

NLP FUNDAMENTALS

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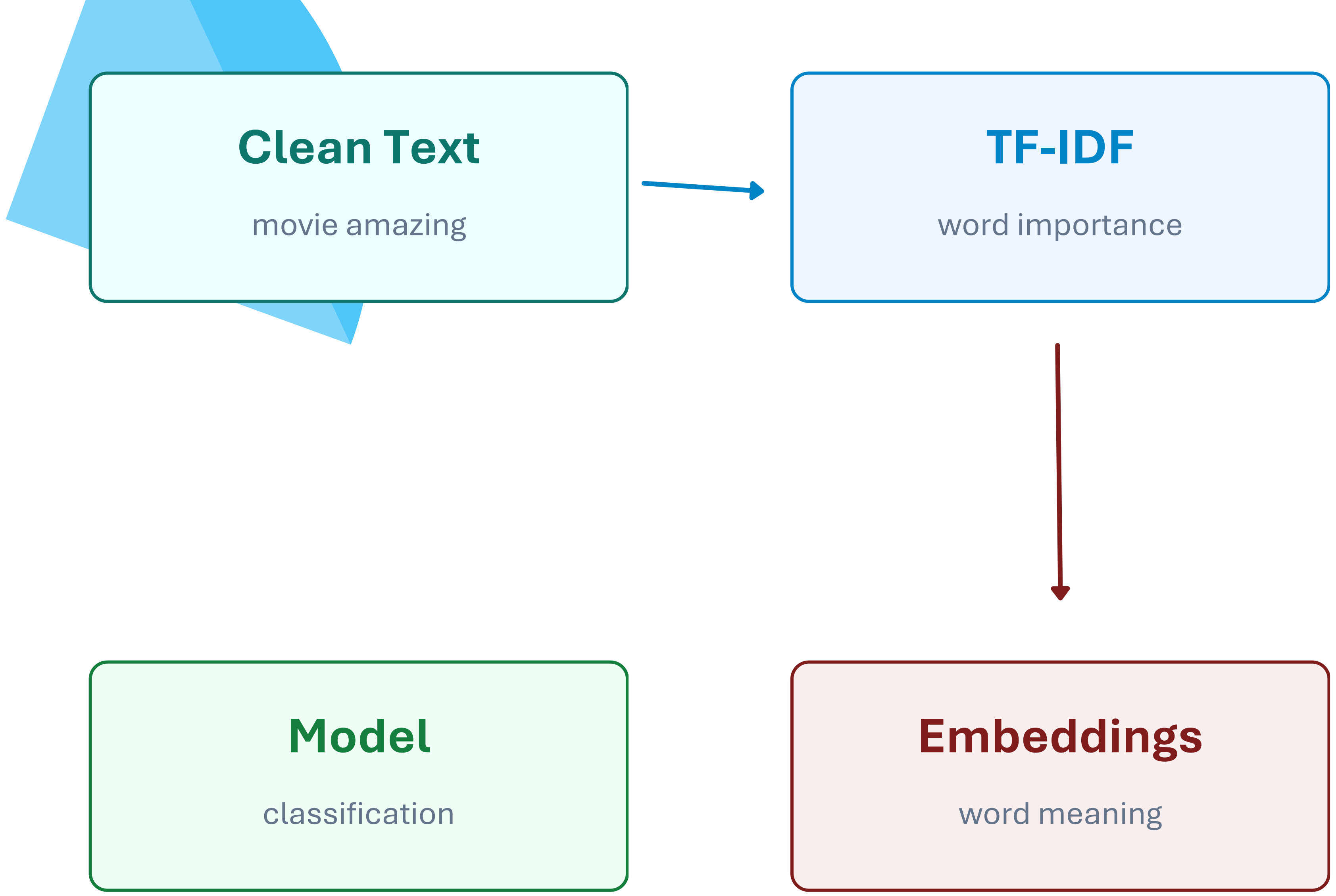
The Day 2 story



One day. Two representation families. One goal: make text usable by machine learning.

1. The Big Picture

Day 1 prepared the text. Day 2 represents it numerically.

Raw Review	→	Clean Text	→	Numeric Vector	→	Model	→	Prediction
"The movie was AMAZING!!!"		movie amazing		[0.0, 0.71, 0.36, ...]		learns patterns		positive / negative

Two ways to represent clean text

TF-IDF

How important is this word in this document?

Frequency-based • sparse • strong baseline

Embeddings

What does this word mean, and what is it close to?

Semantic • dense • captures relationships

Memory sentence: TF-IDF represents importance. Embeddings represent meaning.

2. Bag of Words

The simplest way to turn text into a vector: count words.

Example documents

- **D1** movie good
- **D2** movie bad
- **D3** movie movie good

→

Vocabulary

- bad
- good
- movie

→

	bad	good	movie
D1	0	1	1
D2	1	0	1
D3	0	1	**2**

Bag-of-Words Vector

Bag of Words keeps counts — but ignores word order and meaning.

Limitation

“dog bites man” and “man bites dog” can look identical if they contain the same word counts.

3. TF and IDF

Two questions that lead us to a better weighting system.

TF Term Frequency

How frequent is the word inside this document?

movie good good amazing

movie	1
good	2
amazing	1

IDF Inverse Document Frequency

How rare or common is the word across all documents?

Common everywhere

movie — Low IDF

Rare / distinctive

masterpiece — High IDF

TF looks locally. IDF looks globally.

4. TF-IDF

Combine local frequency with global rarity.

TF-IDF = TF × IDF → word importance score

Why "movie" is down-weighted

movie — TF: high — IDF: low

Appears often in movie reviews, so it does not distinguish one review very well.

Final weight: lower

Why "masterpiece" stands out

masterpiece — TF: moderate — IDF: high

Appears in fewer reviews, so it carries more distinctive information.

Final weight: higher

High TF-IDF ≈ frequent here + uncommon everywhere else.

5. TF-IDF Matrix & Sparse Representation

What the model actually receives.

Vocabulary → Features

- acting
- amazing
- bad
- movie
- story
- wonderful

Each term becomes one column.

TF-IDF Matrix

	acting	amazing	bad	movie	story	...
R1	0	0.71	0	0.08	0.32	...
R2	0.21	0	0.66	0.09	0	...
R3	0	0.54	0	0.07	0.41	...
R4	0	0	0.73	0.10	0	...

Rows = reviews • Columns = terms • Values = TF-IDF scores

Sparse = most values are 0

Because one review uses only a small fraction of the full vocabulary.

6. Word Embeddings

Move from “importance” to “meaning”.

TF-IDF

One document → sparse feature vector

[0, 0, 0.62, 0, 0, 0.18, 0, ...]

Question answered:

How important is each word in this document?

Sparse and frequency-based

Embedding

One word → dense semantic vector

good → [0.21, -0.43, 0.87, 0.15, ...]

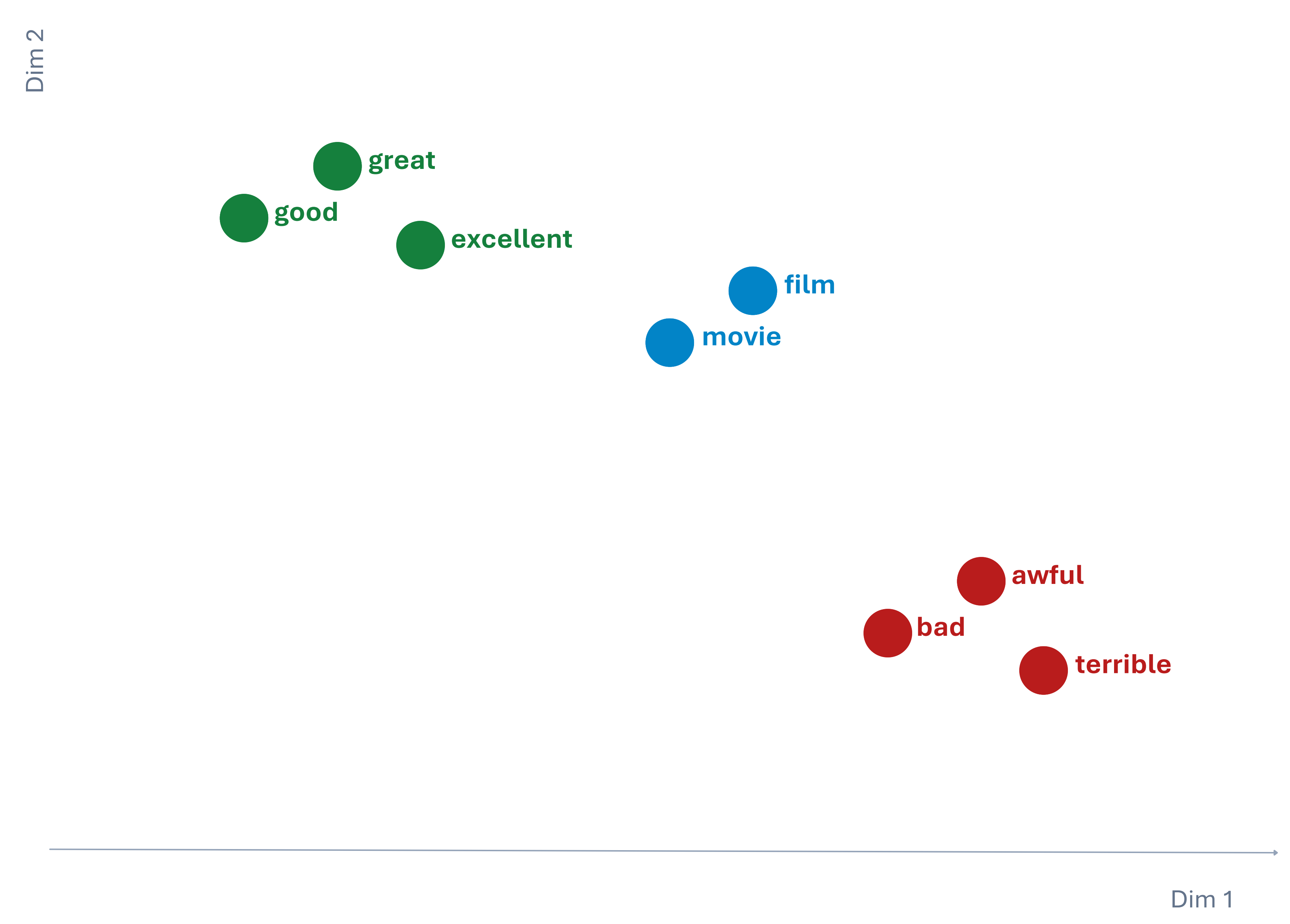
Question answered:

What does the word mean, and which words are related?

Dense and semantic

7. Vector Space & Semantic Similarity

Similar meanings should live close together.



Semantic similarity

good ≈ great ≈ excellent

movie ≈ film

Classic intuition

king − man + woman ≈ queen

Geometry can encode relationships, not just counts.

8. Word2Vec, GloVe & Contextual Embeddings

Three names — one progression toward richer meaning.

01

Word2Vec

Learns dense word vectors from the context in which words appear.

Classic word embedding

02

GloVe

Learns vector relationships using global word co-occurrence information.

Classic word embedding

03

Contextual

Produces a word vector that can change depending on the sentence.

Transformer-style embedding

Why context matters

I deposited money in the **bank** → **financial meaning**

We sat beside the river **bank** → **river-side meaning**

9. TF-IDF vs. Embeddings

Choose the representation based on the question you need to answer.

Question	TF-IDF	Embeddings
Represents	Word importance	Word meaning
Main idea	Frequency + rarity	Semantic geometry
Vector type	Sparse	Dense
Similar words?	Not directly	Yes
Best role	Fast, strong baseline	Semantic / deep learning tasks

Rule of thumb: start with TF-IDF for a clean, fast baseline; use embeddings when semantic meaning matters more deeply.

10. Hands-On Lab — Day 2

What we actually do with the cleaned IMDb reviews.

STEP 1
TF-IDF Baseline

Clean reviews → TF-IDF → simple classifier → evaluate accuracy

STEP 2
Pre-trained Embeddings

Load embeddings → choose a word → find nearest semantic neighbors

STEP 3
Compare Representations

TF-IDF model vs. LSTM / transformer on the same metric

STEP 4
Document the Choice

Which representation fits the project best — and why?

> Lab mindset: do not just run code — explain what each representation is giving the model.

11. Day 2 Mental Map

One slide to remember the whole day.

CLEAN TEXT

How do we represent it?

TF-IDF

TF × IDF

Importance • sparse • baseline

Embeddings

Word → dense vector

Meaning • similarity • context

> **Golden rule: Day 1 prepares the text. Day 2 turns the text into numerical representations.**